

Immunological products

Content

- Introduction
- Vaccine

Learning objectives

- At the end of this session, the students will be able to:
 - ✓ differentiate between biopharmaceuticals and biological products.
 - ✓ describe the types of vaccines
 - ✓ Describe the advantage and drawback of each vaccine type
 - ✓ Discuss immunological events happening after vaccine administration
 - ✓ Define adjuvant ,delivery system and their mechanisms,
 - ✓ Describe general method used to manufacture vaccine
 - ✓ Understand vaccine transportation system

Biopharmaceuticals/biotech drugs

Biopharmaceuticals (biotech drugs)

- drugs produced in living systems by applying bio-engineering/biotechnology.
- Include therapeutic peptides, proteins, antibodies, oligonucleotides, etc.

Biologics/biologicals

- *Biologics (or biologicals) products* are medicinal products produced in living systems such as microorganisms, animal cells or plant cell .
- Biologics not only include biopharmaceuticals, but also whole blood and blood components, vaccines, antibodies and other biomolecules extracted directly from natural (non-engineered) sources.

Biologics/biologicals

- Could be made of sugars, proteins or nucleic acids or complex combination of these substances or may be living entities such as cell and tissues
- Like drugs biological products are used to treat or cure medical condition, prevent disease or diagnose diseases.
- Are made from a variety of natural sources.

Immunological products

- **Immunological products** are preparations that are administered into a live host to induce *artificial acquired immunity*.
 - ✓ Vaccines induce active immunity
 - ✓ Antibodies induce passive immunity

Conventional immunological and biological products

- Conventional immunological and biological products includes:
 - ✓ vaccine
 - ✓ Antibody preparations derived from blood and human plasma
 - ✓ Hormone preparations isolated from biological fluids
 - ✓ Therapeutic enzymes isolated from biological fluids, such as urokinase from urine

The Immune system

- The body is protected from infectious agents and other harmful substances by a variety of cells and molecules that make up the immune system.

Antigens : molecules from a pathogen, cancer cell or foreign substance that provoke a specific immune response.

The Immune system.....

- Foreign substances such as viruses, bacteria, toxins, and parasites are surrounded by antigens that, when introduced into the body, are **capable of inducing a response by the immune system.**
- **This discriminatory ability provides protection from infectious disease, since most agents or associated toxins are identified as foreign by the immune system.**
- **The immune system depends upon the activities of three categories of white blood cells (WBCs) that are derived from bone marrow**

The Immune system.....

Phagocytic cells:

- **Macrophages and dendritic cells** are phagocytic cells that reside in the **blood and tissues waiting to engulf foreign substances.**

T cells:

- After leaving the bone marrow, some WBCs reach the thymus gland where they differentiate and become thymus-derived lymphocytes or T cells.

B cells:

- WBCs that do not reach the thymus gland become B lymphocytes or B cells

Immunity

- **Immunity** is the ability of the human body to tolerate the presence of material indigenous to the body (self), and to eliminate foreign (non-self) material.

Types of immunity

- **Passive**
- **Active**
 - Innate
 - Adaptive

Types of Immunity

Passive Immunity:

- Antibodies from another person or animal that can be injected or transfused.
- Called passive because the individual did not create the antibodies, but instead received pre-formed antibodies.
- Protection is effective, but duration is short lived and no memory is created.
- Examples of passive immunity are
 - maternal antibodies (trans-placental and breast milk) and
 - injected antibodies (e.g., rabies and tetanus immune globulins).

Types of immunity.....

Active Immunity:

- When the body is exposed to a foreign substance the cells of the immune system “actively” respond.
- It can be due to natural infection or vaccine.
- Active immunity is further divided into 2 categories:
 - ✓ **Innate Immunity** - protective mechanisms we are born with
 - ✓ **Adaptive Immunity(Acquired)** – cell mediated immunity and humoral immunity

Types of immunity.....

Innate immunity

- Consists of protective mechanisms we are born with, and are the first line of defence against anything recognized as non-self.
- The produced immune response is **not specific to the antigen and no memory of the antigen persists.**
- However, innate immunity is the crucial first step in most adaptive immune responses.

Types of immunity.....

The following are the protective mechanisms of
Innate immunity

- Physical and Chemical Mechanisms
- Phagocytosis
- Molecular Response
- Inflammatory Response

Types of immunity.....

Innate immunity

Physical and Chemical Mechanisms

Physical barriers:

- Intact skin
- Mucous membrane barrier (sneezing, coughing)
- Cilia

Chemical barriers:

- Tears
- Acid (pH)
- Saliva

Types of immunity.....

Innate immunity

Phagocytosis

Macrophages:

- Engulf and kill invading organisms

Dendritic cells:

- Engulf pathogen
- Display antigen on cell surface
- Travel to lymph node to present antigen to T cells
- Critical link between the **innate and adaptive immune responses.**

Types of immunity.....

Innate immunity

Molecular Response

Cytokines:

- Cytokines are **small proteins** made by a cell that affect the behavior of other cells.
Examples: Cytokines cause **vasodilatation** (heat and redness).
- Some types of **Interferon** are antiviral cytokines which help healthy cells resist viral infection

Chemokines:

- Chemokines are **proteins secreted by macrophages that attract cells out of the blood stream and into the infected tissues.**

Complement system

- Is a group of approximately 20 proteins that coat bacterial surfaces and **promote bacterial destruction by macrophages .**

Types of immunity.....

Innate immunity

Inflammatory Response

- The accumulation of fluid and cells at the site of infection causes the redness, swelling, heat, and pain **known as inflammation.**

Inflammation is beneficial because it:

- recruits cells out of the blood stream,
- increases the flow of lymph to take away microbes and antigen-bearing cells to the lymphoid tissue which will lead to adaptive immunity, and
- brings the T cells and B cells back to the site of infection.

Types of immunity.....

Adaptive immunity

- is the second line of defence against **anything recognized as non-self** and it provides **protection against re-exposure to the same pathogen.**

Characteristics of adaptive immunity:

- **Specificity:** the immune response is specific to the antigen that produced it (e.g. antibody for measles antigen has no effect on rubella antigen)
- **Tolerance:** the immune response is able to differentiate between self and non-self so that body tissues are not destroyed.
- **Memory:** with subsequent exposure to an antigen there is a rapid and strong immune response. This is called an **anamnestic** response.

Types of immunity.....

Adaptive immunity is divided into two categories:

✓ **Cell mediated immunity and Humoral (antibody) immunity**

- **Cell mediated immunity** describes any immune response where **T cells** have the main role.
- T cells **do not recognize** microorganisms in the **extracellular fluids**. Instead, T cell receptors bind to fragments of antigens (epitopes) that are presented on the surface of antigen presenting cells (APC).
- APC carry receptors on their surface **pattern-recognition receptors (PRRs)** that recognize common features of pathogens called **pathogen associated molecular patterns (PAMPs)**.

There are three main types of APC: Macrophages ,Dendritic cells

- Antigen binding B cells
-

Types of immunity.....

- When T cells recognize an antigen presented by the APC, they can differentiate into several different types of T cells:
 - ❖ **Cytotoxic T cells:**
 - Kill cells infected with intracellular pathogens such as viruses
 - ❖ **Helper T cells:**
 - Stimulate B cells to differentiate and produce antibodies.
 - Activate macrophages to become more efficient at killing the pathogen

Types of immunity.....

- **B cells are not activated by most antigens without help from helper T cells.** The activation of T cells is an essential first stage in virtually all adaptive immune responses. This is called the “**T cell-dependant immune response**”.
- Control intracellular bacterial infections (e.g. tuberculosis) that grow in intracellular membrane-bound vesicles of macrophages.
- ❖ **Regulatory T cells:**
 - Suppress lymphocytes and control the immune response
 - Maintain tolerance to self antigen and prevent autoimmune diseases

Types of immunity.....

Humoral immunity

- is mediated by B cells. B cells react against foreign substances in the extracellular spaces of the body by producing and secreting antibodies (Abs).
- These Abs are present in the biological fluids of the body (the humours); hence the term humoral immunity.
- Many microorganisms multiply in the extracellular spaces of the body, and **most intracellular pathogens spread by moving from cell to cell through the extracellular fluids.**
- **These extracellular spaces are protected by humoral immunity where antibodies either kill the extracellular organism and the intracellular organism as it is moving from cell to cell or bind the pathogen and present it to T cells.**

Types of immunity.....

- B cells display immunoglobulin molecules (antibodies) on their surface membranes, which act as receptors for the antigens.
- B cell antibody receptors can either bind to helper T cells that have interacted with an APC or bind to extracellular microorganisms such as bacteria.
- Once an antigen binds to an antibody with the best “fit”, the B cell differentiates into plasma cells or B memory cells.

Plasma cells:

- These cells operate as factories to manufacture the chosen antibody and then secrete those antibodies.

B memory cells:

- These cells mediate immunological memory. They respond rapidly on re-exposure to the antigen that originally induced them.

Types of immunity.....

T cell-dependent antigens

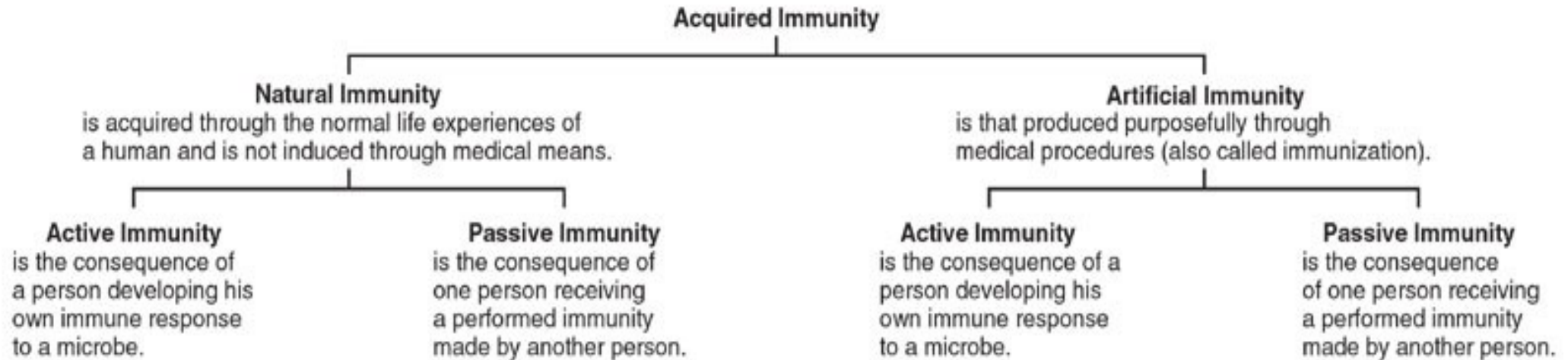
- Most antigens require the interaction of T cells and B cells to generate the production of antibodies. These antigens are referred to as **T cell-dependent antigens**.
- The antibodies produced in response to T cell-dependent antigens are primarily IgG and the response produces **immunologic memory**.

Types of immunity.....

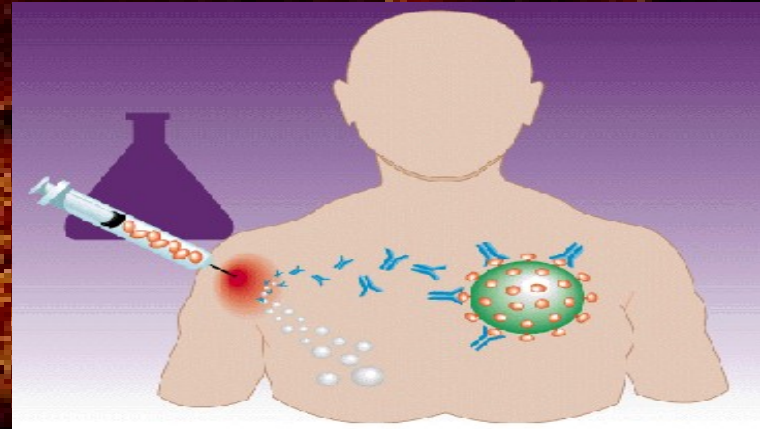
T cell-independent antigens

- In some situations, B cells can create antibodies without the help of T cells.
- Many common extracellular bacteria (e.g. *Haemophilus influenzae* type b) are surrounded by a polysaccharide capsule that enables them to resist ingestion by phagocytes and **therefore avoid stimulating the T cell response**
- Antibodies produced are of the IgM class and immunologic memory is not created.

Summary



VACCINES



What are vaccines?

- A vaccine is an antigenic material that stimulate adaptive immunity to a disease.
- Vaccines are generally considered to be the most effective method of preventing infectious diseases.

Ideal properties of vaccines

- ✓ Should be antigenic but not pathogenic
- ✓ Should have low level of adverse side effects or toxicity
- ✓ Should protect against exposure to natural, wild type forms of pathogen
- ✓ Should stimulate both antibody (B-cell) and cytotoxic (T-cell) responses

Ideal properties of vaccines.....

- ✓ Should induce long lasting immunological memory
- ✓ Should not require numerous doses or boosters
- ✓ Should be cost effective and easy to administer
- ✓ Should be stable, and have long shelf life

Types of Vaccines

- **Live-attenuated vaccines**
- **Inactivated (killed) vaccines**
- **Subunit vaccines**
 - ✓ Toxoids
 - ✓ Polysaccharide
 - ✓ Congugated vaccine
 - ✓ Recombinant vaccines
- **Vector Vaccine**
- **mRNA vaccine**

1. Live-attenuated vaccines

- ✓ Virulent pathogenic microorganisms are treated to become attenuated or weakened without devoid of their antigenic characteristics.
- ✓ Achieved by growth under abnormal culture conditions
- ✓ They have lost their pathogenicity but retain their immunogenicity.
- ✓ These vaccines should not be administered to Immuno compromised individuals.

1. Live-attenuated vaccines....

Advantages as vaccines over non-living vaccines.

- After administration, live vaccines may replicate in the host similar to their pathogenic counterparts.
- This confronts the host with a larger and more sustained dose of antigen, which means that few and low doses are required.
- **In general, the vaccines give long-lasting humoral and cell-mediated immunity.**

1. Live-attenuated vaccines....

Live vaccines drawbacks.

- Live viral vaccines bear the risk that the nucleic acid is incorporated into the host's genome.
- **Reversion to a virulent form may occur**
- It is labile can be damaged and destroyed by heat and light so it should be handled carefully.

1. Live-attenuated vaccines...

- Examples of live-attenuated vaccines include:
 - **BCG (Bacille Calmette Guerin)**
 - Typhoid (oral)
 - **Sabin polio (Oral)**
 - Yellow fever
 - **Measles**
 - Mumps
 - Rubella (German Measeles)
 - Intranasal influenza
 - Typhus

2. Inactivated (killed) vaccines

- ✓ Pathogenic microorganisms are killed or inactivated by heat, chemicals (e.g., formaldehyde) or radiation without devoid of their antigenicity.
- These vaccines are **usually safe but less effective than live-attenuated** vaccines.
- **It requires multiple dose**
- The immune response to inactivated vaccine is **mostly humoral, little or no cellular immunity results.**
- Antibody titers against inactivated antigens diminish with time. As a result some inactivated vaccines may require **periodic supplemental dose to increase or bust antibody titers**

2. Inactivated (or killed) vaccines.....

- Examples of killed or inactivated vaccines:
 - ✓ Typhoid
 - ✓ **Cholera**
 - ✓ **Pertussis**
 - ✓ Salk polio (injection)
 - ✓ **Rabies**
 - ✓ Japanese encephalitis

Attenuated vs killed vaccine

COMPARISON OF ATTENUATED (LIVE) AND INACTIVATED (KILLED) VACCINES

Characteristic	Attenuated vaccine	Inactivated vaccine
Production	Selection for avirulent organisms: virulent pathogen is grown under adverse culture conditions or prolonged passage of a virulent human pathogen through different hosts	Virulent pathogen is inactivated by chemicals or irradiation with γ -rays
Booster requirement	Generally requires only a single booster	Requires multiple boosters
Relative stability	Less stable	More stable (advantageous for Third World countries where refrigeration is limited)
Type of immunity induced	Produces humoral and cell-mediated immunity	Produces mainly humoral immunity
Reversion tendency	May revert to virulent form	Cannot revert to virulent form

3. Subunit vaccines

- Subunit vaccines comprise one or more components of a pathogen rather than the entire pathogen.
- It involves isolation of a certain specific protein from a virus or bacteria and administering this by itself.
- **A weakness of this technique is that isolated protein may have a different three dimensional structure than the protein in its normal context and will induce antibodies that may not recognize the infectious organism.**
- Have weaker antibody response than the other classes.
- Considered to be much safer than **their attenuated counterparts.**

3. Subunit vaccines.....

- **The most common subunit vaccines include:**
- ✓ **Toxoids**
- ✓ **Polysaccharides**
- ✓ **Conjugate vaccines**
- ✓ **recombinant vaccine**

3. Subunit vaccines.....

Toxoids

- Toxoids are prepared by detoxifying the exotoxins of some bacteria rendering them antigenic but not pathogenic.
- **Antibodies produced in the body as a consequence of toxoid administration neutralize the toxic moiety produced during infection rather than act upon the microorganism itself.**
- Examples of toxoid-based vaccines include **diphtheria toxoid** (*Corynebacterium diphtheriae*) and **tetanus toxoid** (*Clostridium tetani*).

3.Subunit vaccines.....

Polysaccharide vaccines

- Contains the polysaccharide coats or capsules of encapsulated organism which are purified and non-infectious.
- They are safe.
- Examples:
 - ✓ **Meningococcal vaccine** from the polysaccharide antigen of the bacterial cell wall (*Neisseria meningitidis*).
 - ✓ **Pneumococcal vaccine** from the polysaccharide contained in the capsule of the bacterium (*Streptococcus pneumoniae*)

3.Subunit vaccine.....

Conjugated vaccine

- Children under 2 years of age do not respond to antigens such as polysaccharide, which produce **antibodies via a T-cell independent mechanism.**

If these polysaccharide antigens are conjugated to a protein that T cells recognize ,then these conjugate can elicit strong immune response and immune memory in young children.

- Eg meningitis A and B conjugate vaccine
- Homophiles influenza type b(Hib) conjugate vaccine

3.Subunit vaccine.....

Recombinant vaccine

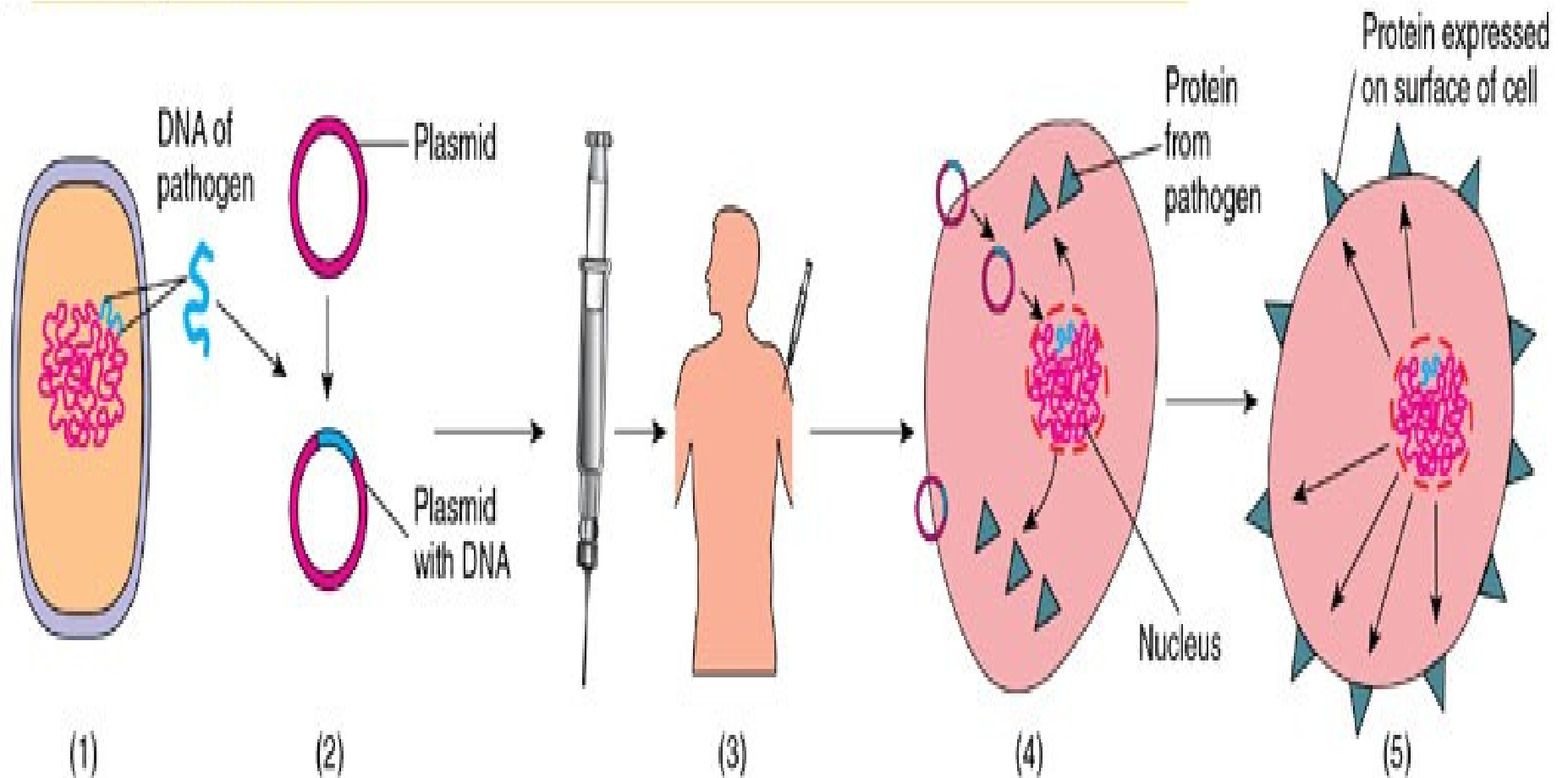
- To improve the yield, facilitate the production, and/or improve the safety of protein-based vaccines, protein antigens are sometimes expressed by host cells of the same (homologous) species or of different (heterologous) species that are safe to handle and/or allow high expression levels.
- Heterologous hosts used for the expression of immunogenic proteins include yeasts, bacteria, and mammalian cell lines.

3.Subunit vaccine.....

- A viral vaccine can be made by inserting a segment surface virus gene into an expression host.
- The modified host cell produce large amount of the virus surface antigen which is purified and harvested and used to produce the vaccine.
- The recombinant vaccine is identical to the natural virus surface antigen but does not contain virus DNA and is unable to produce infection.

4.Vector Vaccines

- ✓ Insert desired gene into a vector
- ✓ Inject a vector into a patient and allow to replicate
- ✓ DNA injections can transduce cells so antigens are expressed and presented



DNA that codes for protein antigen extracted from pathogen genome.

Genomic DNA inserted into plasmid vector; plasmid is amplified and prepared as vaccine.

DNA vaccine injected into subject.

Cells of subject accept plasmid with pathogen's DNA. DNA is transcribed and translated into various proteins.

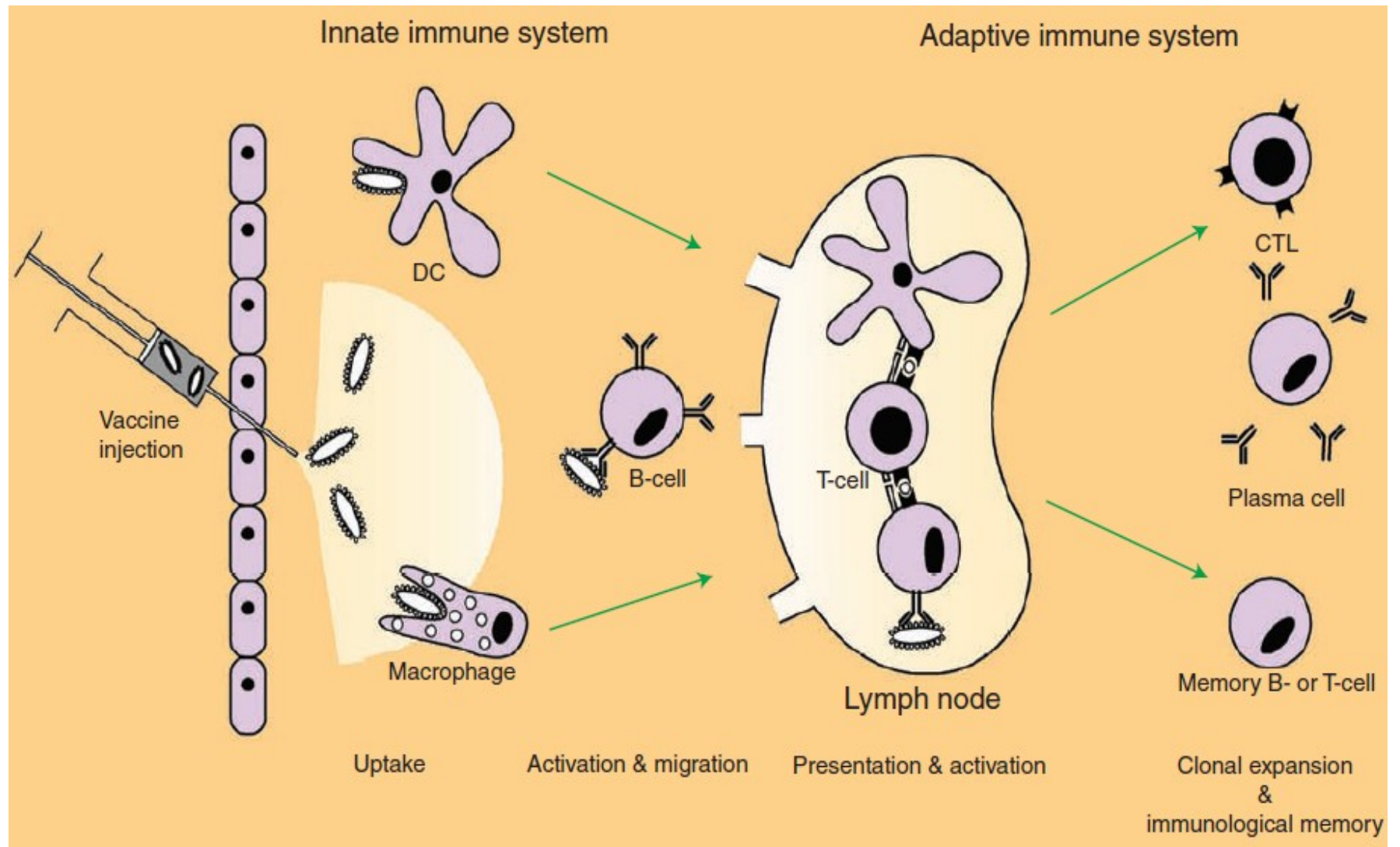
Foreign protein of pathogen is inserted into cell membrane, where it will stimulate immune response.

5. mRNA vaccine

Eg. Pfizer covid-19 vaccine

Events after vaccine administration

- ✓ Uptake of the vaccine by phagocytic cells.
- ✓ Activation and migration of professional antigen-presenting cells (APCs) from infected tissue to peripheral lymphoid organs.
- ✓ Antigen presentation to T helper-lymphocytes.
- ✓ Activation of pathogen specific T- and B-lymphocytes.
- ✓ Formation of effector cells and memory cells of T and B-lymphocytes.
- ✓ Revert of memory cells to inactive state.



Adjuvants and delivery system

- ✓ The term “adjuvant”, derived from the Latin *adjuvare* which means to help.
- ✓ It has been used to describe any material that can increase the humoral or cellular immune response to an antigen.
- ✓ Adjuvants can improve the potency of vaccines when they are formulated with the antigens **but do not confer immunity on their own.**

Adjuvants....

Adjuvants and delivery systems stimulate the immune system by several mechanisms of action

- A depot effect leading to slow antigen release and prolonged antigen presentation.
- Attraction and stimulation of APCs by some local tissue damage and binding to PRRs present on APCs.
- Delivery of the antigen to regional lymph nodes by improved antigen uptake, transport and presentation by APCs(carrier).

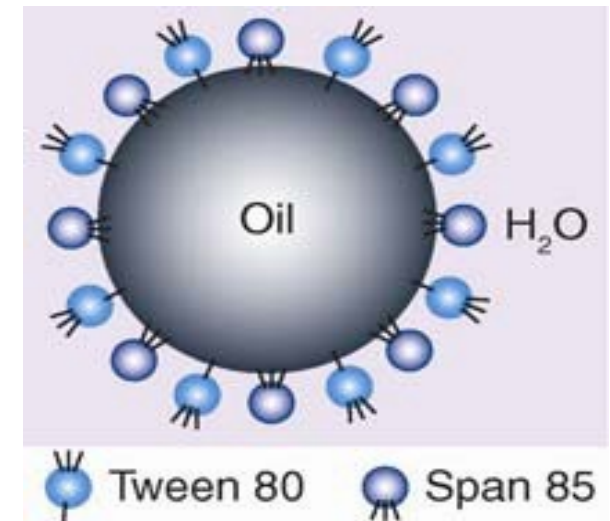
Adjuvants....

- The adjuvants include:
 - **Aluminium salts** ~ Alum (or Aluminium sulphate), Aluminium hydroxide, Aluminium phosphate.
- These are FDA approved adjuvants for use in humans.
- The antigen is adsorbed on aluminium salts.
- The antigen is slowly released from the insoluble salt particles, which allowed prolonged and effective stimulation of the immune system. .

Adjuvants....

□ Freund's Emulsions

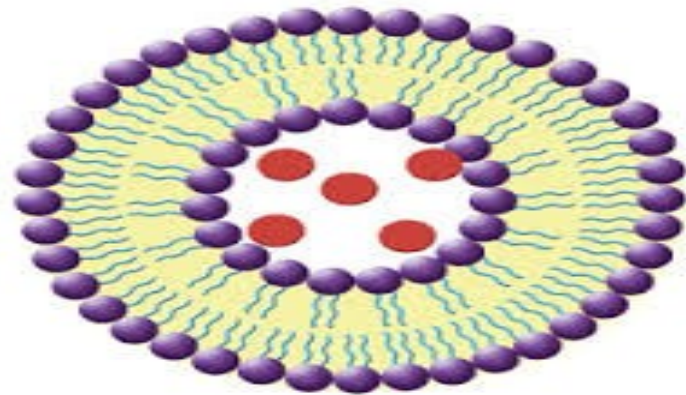
- Freund's adjuvant is water-in-oil emulsion of mineral oil and water stabilized by emulsifiers.
- This seems to act by stimulating antibody-forming cells mainly at the site of injection.
- Emulsion adjuvants promote or enhance T cell response.



Adjuvants....

□ Nanoparticles

- A nanoparticle is a submicroscopic solid particle with size ranging from 10 nm to 1 μm .
- The size of nanoparticle allows it to be administered intravenously with little risk of embolism.



Adjuvants....

- Sterilizable, non-toxic and biodegradable.
- Examples of materials for preparation of nanoparticles include albumin, gelatine materials used in the preparation of nanoparticles are, starch, chitosan, poly lactide-co-glycolide, etc.

General steps in manufacturing of vaccines

Growth of the pathogen

(e.g., propagation in embryonated chicken egg, monkey kidney cells, artificial media, etc)



Isolation of pathogen or cellular component

(e.g., centrifugation, lysing cells to form suspension, etc)



purification (e.g., precipitation, ultrafiltration, column chromatography, etc)



Formulation, filling and packaging

(freeze drying, co-precipitation with adjuvants, addition of (stabilizers, preservatives, etc)

Transportation and storage of vaccines

- The immunogens in vaccines are susceptible to alteration or inactivation by heat, freezing and extremes of pH.
- Thus, they should be handled in a “cold chain” system during transportation and storage.
- The “cold chain” is a temperature-controlled supply chain commonly used in food and pharmaceutical industries

Transportation, storage and administration....

- Most vaccines should be stored at refrigerator temperatures (2- 8 °C) but a few are frozen (e.g, chickenpox vaccine) and some for field use may not require refrigeration.
- **The use of pH indicators or temperature or time sensitive labels (“vial vaccine monitors,”) which change color when exposed to extreme temperatures or after the expiration date)** can avoid unintentional administration of inappropriately stored or expired vaccine.

■

Manage vaccine carefully!!!